

Project Information

Title: Object Pattern Recognizer

Team: 3 developers/team preferred, single work on request

Introduction

In this project, you will gain hands-on experience by developing a small Python application that uses the livestream of a camera and performs real-time object pattern recognition and color detection. The recognition is visualized within the livestream and the information gathered is logged into a csv file for later analysis. The goal is to learn the basics of software development including following aspect:

- **Dependency management:** Gain experience in handling external libraries and packages
- **Architecture and Code Structuring:** Develop a well-structured and modular codebase that uses to best practices in software architecture.
- **Documentation:** Improving your skills in writing clear documentation including static and runtime-view diagrams.
- **Version control systems:** Familiarize yourself with version control systems like git and start working with issues, pull requests co.

Must haves

Features

- **Object Pattern Recognition:** The system should be able to detect patterns like circle, rectangle, square, triangle.
- **Color Detection:** Inside each pattern, the system shall determine the main color of the pattern. Following colors should be detectable: red, green, blue, yellow, violet.
- **Real-time Visualization:** Overlay recognized objects with their respective patterns and colors on the live video/image.
- **Data Logging:** Log timestamp, pattern type, detected color, and any other relevant information into a structured CSV file.
- **System operation types:** The system shall support two different types of operation. This shall be solved by using command line arguments.
 - Type **CAMERA**: detects shapes and colors using your notebook camera live-stream
 - Type **IMAGE**: analyzes all images in a given folder-path
- **System configuration:** The whole system configuration (.e.g path to log-folder, thresholds and co.) shall be stored in a separate file. ([hint](#))

Documentation and others

- **Dependency Management:** Solve the dependency management problem, so everybody can run the code with ease.
- **Version Control System:** Use a version control system like git (in combination with gitlab) to ensure proper version and release controlling. Use issues, merge requests and feature branches with peer-to-peer reviews.

- **Clean Code Principles:** Use of advised clean code principles while writing your code.
- **Architecture:** Organize your codebase into modular components for maintainability and extensibility. Create classes/modules with distinct responsibility. Create a component diagram with all components and their relation to each other. The architecture shall be a part of the repository.
- **Detailed Documentation:** Create documentation to guide users through your applications setup, features, and usage. Please provide a meaningful runtime view diagram showing what your application does. The documentation shall be a part of the repository.

Optional features

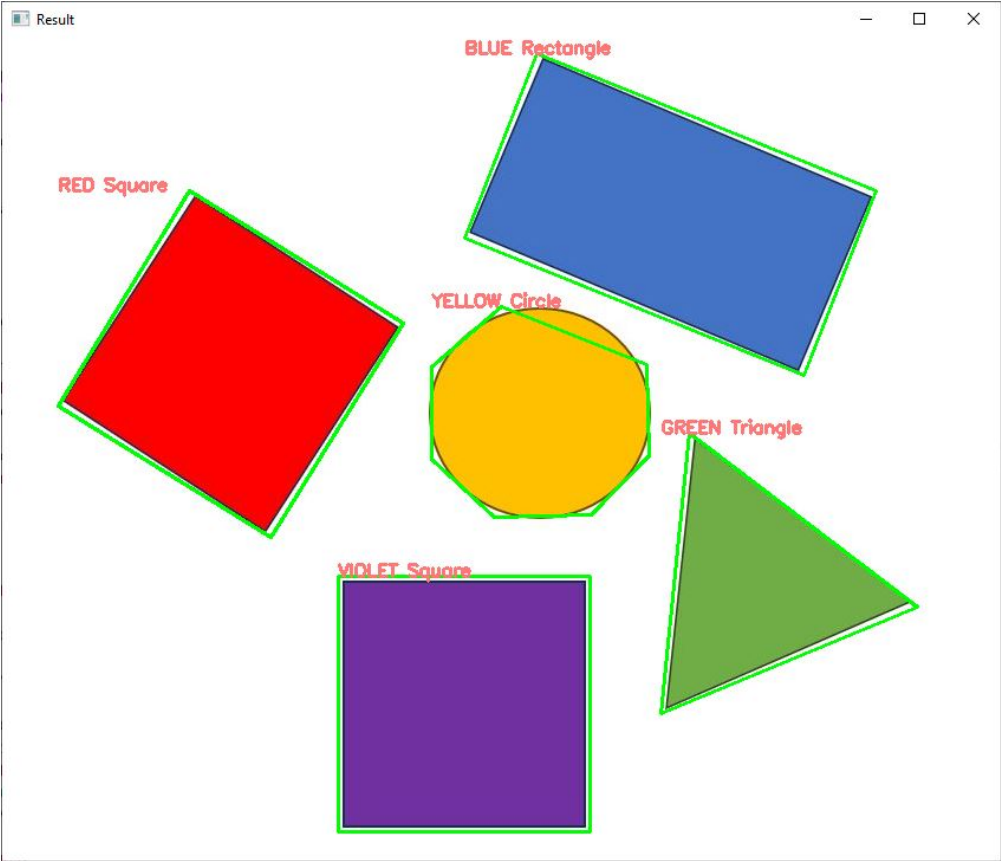
To get a better grade, following features can be implemented:

- **GUI:** Create a GUI using pyqt. You're free how it should look like.
- **FEAT:** Acoustic shape and color output (you can use libraries like gTTS)

Examples

Pattern and color recognition

The provided sample image can be used for testing purposes.



Logging

Timestamp	Pattern	Color
2023-08-28 17:12:47.090492	Square	VIOLET

	2023-08-28 17:12:47.093486		Triangle		GREEN	
	-----		-----		-----	
	2023-08-28 17:12:47.094483		Circle		YELLOW	
	-----		-----		-----	
	2023-08-28 17:12:47.096480		Square		RED	
	-----		-----		-----	
	2023-08-28 17:12:47.098484		Rectangle		BLUE	
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Project Submission

- Create a **tag** of your repository (e.g. v1.0.0)
- Export the **tag** content as a **zip**
- Create a **txt file** containing the link to your **public repository** and a list of additionally developed features
- Upload the **txt file** including the exported **zip** to the moodle submission form
- done

Questions or support needed?

You can always contact me using this e-mail address: klims.saikins@cedes.com
I will try to respond inner 24 hours.